



Stochastics and Statistics

Dynamic advance scheduling of outpatient appointments in a moving booking window

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ARTICLE INFO

Article history:

Received 20 February 2020

Accepted 19 November 2020

Available online 27 November 2020

Keywords:

OR in health services

Advance scheduling

Markov decision process

Booking window

Discrete convex analysis

ABSTRACT

This paper addresses the dynamic advance scheduling for outpatient medical appointments in a moving booking window. To address this problem, a finite-horizon Markov decision process (MDP) model is proposed to maximize the total expected net reward comprising the reward for accepting outpatients and the capacity usage cost. Structural properties of the optimal value function and the optimal control policy are established. For the special cases in which the lengths of the booking window are equal to 1 and to 2 periods, the antimultimodularity of the optimal value function is proved while the monotonicity and bounded sensitivity of the optimal control policy are established. For the general cases, partial characterization of the optimal policy is performed. Based on insights from the theoretical results, two efficient heuristic policies are devised, including the assignment-sequence policy and multiple-threshold policy. Numerical experiments show that both policies perform consistently well and outperform the other benchmarking heuristic policies.

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1. Introduction

Diagnostic facilities are a critical resource in hospitals, given the associated high investments and operating costs. As a result, they are generally shared by multiple medical units such as the emergency department, hospital wards, and different outpatient departments (Patrick, Puterman, & Queyranne, 2008). Emergency patients usually present as urgent cases and require same-day examinations, whereas outpatients are non-urgent and can be served after a lag time of days or even weeks. In the appointment-scheduling of outpatient imaging examinations, hospital managers usually aim at effectively reducing patients' waiting time while also reserving capacity for emergency cases. However, reserving too much capacity leads to more idleness of critical resources and longer indirect waiting time for the outpatients from the date of reservation to the date of service; conversely, reserving too little capacity leads to overcrowding and overtime of imaging facilities and longer direct waiting time for outpatients from the time of arrival to the time of service. For both outpatients and emergency patients, such resultant long waits could adversely affect the care quality.

Therefore, hospital managers are under pressure to maximize the use of their facilities without compromising their patients' waiting time.

In our collaborating hospital, diagnostic facilities are shared between emergency patients and outpatients. The emergency patients typically need same-day imaging examinations due to the severity of their cases, whereas the outpatients are scheduled according to the first-come-first-served (FCFS) rule within a booking horizon of 4 to 6 weeks. Hospital managers usually first allocate the capacity of the diagnostic facilities between the emergency patients and outpatients, and then serve the outpatients by making appointments in advance and serve the emergency patients by using the same-day dedicated capacity or overtime/extra capacity. This static way is convenient and easy to implement. However, due to dynamic arrivals of emergency patients over different days, it is challenging to devise a good static capacity-allocation decision. Therefore, it is important for hospital managers to make dynamic advance-scheduling decisions, i.e., dynamically scheduling the outpatients to dates within a designated booking horizon or diverting them to other hospitals, with concurrent consideration for potential arrivals of emergency patients.

Motivated by this problem, we propose a finite-horizon Markov decision process (MDP) model for the outpatient advance-scheduling problem with the twofold objective of maximizing the revenue from accepting the outpatients and minimizing the

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capacity usage cost. The resource of interest is divided into multiple time slots (e.g. 15- or 30-minute slots), and each patient requests one time slot. Outpatients randomly arrive in the hospital; the number of the outpatients is assumed to be known at the beginning of each day. Then, schedulers need to make the advance-scheduling decision, i.e., assigning the outpatients to dates within a designated booking horizon, or rejecting them. In practice, the outpatients are classified into multiple priorities, the presence of which complicates the formulation and analyses of the problem due to the curse of dimensionality. Accordingly, this paper considers a simplified setting involving only one type of outpatients. The structural properties of the optimal advance scheduling policy are established. For special cases (in which the booking horizons are 1 and 2 days long), the optimal value function is proved to be discrete jointly concave (antimultimodular). This implies that a local optimum is also a global one. For general cases, bounds and partial characterization of the optimal policy are established. Besides, for both types of cases, the monotonicity, bounded sensitivity, and bounds of the optimal policy are established. With insights from the theoretical results, we propose two main heuristic policies: the assignment-sequence policy (ASP) and multiple-threshold policy (MTP). The ASP makes the advance-scheduling decision based on a predetermined assignment sequence; whereas the MTP does so based on different thresholds or control limits for different periods within the booking window. Numerical experiments are conducted to compare the performances of both heuristic policies, several other commonly-used heuristic policies, and the optimal control policy. The numerical results demonstrate that the ASP and MTP are close to the optimal policy in small-sized cases and consistently outperform the other heuristic policies in real-sized cases.

Note that the mathematical model and the results of this paper are not restrict to medical appointment scheduling and can be applied to appointment scheduling of general service systems such as banks and production scheduling. The major restriction is the assumption of identical service capacity requirement which allows us to define the resource usage cost as a function of the number of appointments for each period. Although this is acceptable for many service systems, it is restrictive for production scheduling.

The remainder of this paper is organized as follows. Following the literature review in Section 2, we describe and formulate the problem in Section 3. In Section 4, we establish analytical results, including the structural properties and bounds of the optimal scheduling policy. Heuristic policies are proposed in Section 5, and numerical experiments are presented in Section 6. Section 7 concludes the paper and provides research perspectives.

2. Literature review

As a widely-studied problem in the literature (Patrick et al., 2008), patient scheduling can be divided into advance scheduling, appointment scheduling, and real-time scheduling (Geng & Xie, 2016). Advance and appointment schedules determine the service date and time interval for regular patients, whereas the real-time schedule determines the service sequence for patients of different priorities. Methodologies that have since been developed include the Markov decision process (MDP), mathematical programming, queuing theory, robust optimization, and dynamic programming. Herein, however, we focus on only the literature on advance scheduling that is most relevant to our paper.

The MDP has frequently been used to formulate dynamic advance-scheduling problems. However, the main barrier is the curse of dimensionality, which makes the MDP models of real-world problems intractable. It is thus crucial to analyze the MDP model and characterize the form of the optimal policies. The theoretical results are then used to design heuristic policies or embedded on approximation schemes such as linear programming

and approximate dynamic programming (ADP). Examples are copious in the literature. An MDP model was proposed in Patrick et al. (2008) to dynamically schedule patients with different priorities to a diagnostic facility in a public healthcare setting with the objective to minimize the total cost. ADP was used to address the model because the state space was overly large for an exact solution. A finite-horizon MDP was proposed in Geng and Xie (2016) to make the advance-scheduling decision. The patient-type-specific maximal allowable waiting time was considered to balance high-priority patients' acceptance rate and capacity utilization. The authors proved the structural properties such as monotonicity and concavity of the optimal reward function and dynamic evolution of the optimal policy. In Gocgun and Ghatge (2012), both allocation and advance-scheduling problems were formulated as MDP models, which were solved by developing an ADP method with Lagrangian relaxation and constraint generation. In Gocgun and Puterman (2014), a dynamic advance-scheduling problem with due dates and time windows was formulated as an MDP, for which a linear-programming-based ADP method was proposed to obtain the approximated policies. Consideration was given in Barz and Rajaram (2015) to elucidating a patient-admission problem with multiple resource constraints, on the assumption that arrivals were random for both emergency and elective patients. In this problem, emergency patients must be accepted whereas elective patients could be accepted, postponed, or even rejected. The control process was accordingly formulated as an MDP model to maximize the expected contribution net of overbooking costs. ADP was used for constructing heuristic policies. A multiple-resource allocation-scheduling problem was formulated in Huh, Liu, and Truong (2013) as an MDP model considering emergency and elective patients. Structural properties were proved, including the convexity of the formulation and monotonicity of the optimal policy. Additionally, novel aspects have received scholarly attention. Besides emergency and elective inpatients, a new type of inpatients, called expedited patients, was considered in Helm, Ahmad-Beygi, and Van Oyen (2011). An MDP model was proposed and the optimal policy characterized for designing heuristic policies. Considering no-shows and lead times in an outpatient appointment-scheduling problem, an MDP model was developed in Patrick (2012). Besides, a dynamic programming model was proposed in Green, Savin, and Wang (2006) combining advance scheduling and real-time scheduling. In Wang, Chen, and Xu (2018), a dynamic programming model was proposed for appointment scheduling considering patient choice.

Mathematical programming for advance scheduling often includes linear programming, mixed-integer linear programming, quadratic programming, etc. These approaches usually focus on static problems, where the dynamics of the system are not considered when decisions are made. Numerous examples are available in the literature. An integer linear-programming model was proposed in Marques, Captivo, and Pato (2012) to schedule elective surgeries from the waiting list on a weekly time horizon with the objective of maximizing the use of the resources in the surgical suite. A similar problem was formulated as a mixed-integer linear programming (MILP) model in Al Hasan, Gu  ret, Lemoine, and Rivreau (2019). In Conforti, Guerriero, and Guido (2010), a scheduling problem for patients on radiotherapy was considered: an integer linear-programming problem was proposed to minimize the mean waiting time or to maximize the number of new scheduled patients. An integer linear-programming model was developed in Adan and Visser (2002) for an admission-planning problem to decide the daily composition of patients, given a target patient throughput and a pre-designated level of resource utilization. A similar problem was considered in Visser, Adan, and Bekkers (2005). Besides the aforesaid approaches, other methodologies have been applied in advance scheduling, including the queuing

theory, stochastic programming, and robust optimization. Several queuing models for formulating hospital admissions scheduling were reviewed in Kolesar (1970). An optimization problem for admissions to an intensive care unit (ICU) was considered in Shmueli, Sprung, and Kaplan (2003). Stochastic programming models were proposed in Kamran, Karimi, and Dellaert (2018) to address an advance scheduling problem in operating room planning. In Zhang, Dridi, and El Moudni (2019), a two-level model for elective surgery scheduling was proposed, in which an MDP model and a stochastic programming model were developed in the first and second level respectively. A distributionally robust optimization approach was proposed in Meng et al. (2015) for managing quotas for elective inpatient admissions.

Within the existing literature, two papers bear relevance to our work: Patrick et al. (2008) and Truong (2015). Accounting for multiple priorities, in Patrick et al. (2008), the authors formulated the scheduling problem as MDP models. The distinctions between the paper and our work are threefold. Firstly, due to the complexity of the MDP model, an approximate model was proposed and analyzed in Patrick et al. (2008) to design ADP algorithms, whereas the original MDP model is analyzed in our paper through discrete convex analysis and the optimal control is characterized to design heuristic policies. Secondly, increasing marginal waiting cost was considered in Patrick et al. (2008) which implies there is no booking window, whereas the booking windows are considered in our paper as hard constraints, due to which patients are either scheduled within the booking window or rejected. Thirdly, keeping patients on the waiting list is allowed in Patrick et al. (2008), whereas in our study the manager must inform the patients about the advance-scheduling decision on the day of arrival. Truong (2015), notwithstanding its relevance to our work, differs significantly by booking any regular patient in the entire remaining time horizon instead of a more realistic moving booking window as in our model. Other differences include rejection and general reward structure in our paper instead of constant waiting cost rate in Truong (2015). These differences make impossible the generalization of analysis technique of Truong (2015).

The main contributions of this work are summarized as follows. Most of the literature on similar problems has concentrated on either proving the structural properties in the continuous state space or directly employing ADP or heuristic policies. We analyze the MDP model in the discrete state space through discrete convex analysis. The structural properties of the optimal control policies are established, including multimodularity and supermodularity of the objective functions and bounds on the optimal policies. Enlightened by the structural properties, we design efficient and easy-to-use heuristic policies for large-scale cases. Lastly, we fill a gap of the theory underlying discrete convex analysis by proving the preservation of multimodularity (antimultimodularity) under minimization (maximization) over the middle part of the list of ordered variables with an extra equality constraint.

3. Problem description and model formulation

In this paper, we consider the problem of scheduling randomly-arriving patients over a moving booking window. Note that emergencies are not explicitly considered but can be easily taken into account as shown at the end of the Section and in Section 6 by appropriate definition of the resource usage cost function. We consider the advance-scheduling decision within a booking horizon, i.e., the assignment of outpatients to a day within the booking horizon and the rejection of patients. Appointment-scheduling decisions, i.e., the assignment of outpatients to a period within the service day, are not considered. More specifically, we consider a discrete-time model with days as the elementary periods. Each day t has a_t new patients and the numbers a_t of different days are as-

sumed to be mutually independent. At the beginning of day t , a_t becomes known and the booking window comprises the current day t plus L subsequent days, i.e., day $t, t+1, \dots, t+L$. The booking state of the existing patients is known. The scheduler decides the number of new patients to schedule for each day of the booking window and the number of new patients to reject. Booking a new patient for day $t+i$ brings a reward r_i , which should be balanced with the capacity usage cost $C_t(y)$ for serving y patients in day t that is assumed to be increasing convex in y . In practice, r_i tends to be non-increasing in i since patients usually prefer a shorter waiting time. If patients exhibit certain preferences of the appointment date, e.g. specific days in a week such as Monday or Friday, the reward pattern can be adjusted accordingly. Although this problem is motivated by the realistic problem with random emergency arrivals, the emergency arrivals are not explicitly incorporated in the formulation, given our attempt to keep the formulation general. In fact, the formulation can easily consider emergency patients by giving the capacity usage cost a specific form (as demonstrated in the numerical experiments in Section 6). The capacity usage cost is also tunable and can consider costs related to aspects such as resource idle time, overtime, and patients' waiting time. In this paper, we consider the last two costs which become more undesirable as they grow longer and, accordingly, the capacity usage cost is assumed to be non-decreasing convex in the number of patients to be served. The capacity usage cost can be easily modified to consider capacity idle time. Idle penalty is ignored for simplicity and clarity.

The problem is comprised of dynamically determining, for each day, the number of new patients to assign to each of the days within the booking window in order to maximize the total expected net reward from accepting patients minus the capacity usage cost thus incurred over a finite time horizon of N days.

The following notations will be used throughout the paper. For clarity, other notations used only within specific analyses or proofs will be introduced later in the corresponding contexts and not presented here.

Notation	Definition
L	length of the booking window
N	length of the planning horizon
r_i	reward for booking a patient to i days later, $i=0,1,\dots,L$
a_t	random number of patient arrivals on day t with $\lambda_t = E[a_t]$
$C_t(y)$	capacity usage cost given that y patients are served on day t , assuming that it is increasing convex in y
x_i	state variable indicating the number of patients booked on i days later, $i=0,1,\dots,L-1$
q_i	decision variable indicating the number of new arrivals to assign to i days later, $i=0,1,\dots,L$
$f_t(\vec{x})$	optimal value function indicating the expected total reward from day t to day N given that the current state is $\vec{x}$

The problem is modeled as a finite-horizon MDP model in the following.

Decision epochs: $T_0=\{0,1,\dots,N\}$: set of decision epochs (days, or equivalently, periods) indexed by t .

State: $\vec{x}=(x_0, x_1, \dots, x_i, \dots, x_{L-1})$: an L -dimensional integer vector, where x_i indicates the number of existing patients booked i days after the current day. For example, x_0 and x_1 respectively denote the number of patients booked on the current day and that on the next day. For convenience, we also use the notation x_L with $x_L=0$.

Action: $\vec{q}=(q_0, q_1, \dots, q_i, \dots, q_L)$: an $(L+1)$ -dimensional integer vector, where q_i indicates the number of new patients assigned to day i of the booking window. For example, q_0 and q_L respectively denote the number of new patients for the current day and that for the last day in the booking window. A natural constraint of the actions is $\sum_{i=0}^L q_i \leq a_t$, where a_t indicates the number of new

arrivals. If the summation is less than a_t , it means that some patients are rejected.

State transition: When the actions are determined, $x_0 + q_0$ are served on the current day and the new booking state of the next day is $\bar{x}_+ = (x_1 + q_1, \dots, x_i + q_i, \dots, x_{L-1} + q_{L-1}, q_L)$.

Rewards: The immediate reward or stage reward is the total reward of new patient booking minus the capacity usage cost and is equal to $\sum_{i=0}^L r_i q_i - C_t(x_0 + q_0)$.

The optimal expected total reward from day t to day N given state $\bar{x}$ is denoted by $f_t(\bar{x})$ and can be determined by the optimality equation

$$f_t(\bar{x}) = E_{a_t} \left[\max_{q_0 + \dots + q_L \leq a_t} \left(\sum_{i=0}^L r_i q_i - C_t(x_0 + q_0) + f_{t+1}(x_1 + q_1, \dots, x_{L-1} + q_{L-1}, q_L) \right) \right] \quad (1)$$

and

$$f_N(\bar{x}) = 0, \forall \bar{x} \quad (2)$$

where N is the end of the scheduling horizon. For the long-run average optimal, N should be made sufficiently large to ensure convergence.

We now address the computational issue of the MDP model. The optimality equation can be solved through backward value iteration. The state space is L -dimensional and the action space is $(L+1)$ -dimensional. The usual state-space truncation is used to accelerate the resolution of the MDP model through the value iteration. However, when L is large, the curse of dimensionality makes the problem impossible to solve. In the following, we first establish structural properties of the optimal scheduling policy for special and general cases in order to gain useful insights and then design heuristic policies for the general case.

The practical relevance of the proposed model is threefold. Firstly, the booking decision for each new arrival is given on the day of arrival. While that is reasonable for most appointment-scheduling problems, it may be beneficial to leave patients on a waiting list and make the booking decisions later, which could be an important open issue for highly-overloaded resources. Secondly, the decision is made upon the knowledge of all new arrivals, which in practice could be realized by making the decision at the end of the day. An alternative is to make the appointment decision one by one and asynchronously upon the arrival of each new patient. Thirdly, the delay-dependent reward r_i is sufficiently flexible to model different realistic situations (the sooner appointments the better or the ideal appointment date due to constraints of other relevant medical procedures). For example, the constant reward $r_i = r, \forall i$ is a special case of interest. The most important is the capacity usage cost $C_t(y)$ which depends only on the number of patients to serve. This implicitly requires that all patients have the same capacity requirement, i.e., either appointment slots of the same duration or identical and independently distributed service times. Another special case of interest concerns the overcapacity usage when a regular capacity H is shared by the appointment patients and a random number ξ of emergency patients, in which case $C_t(y) = E_\xi[(y + \xi - H)^+]$, as considered in Section 6.

4. Structural properties and bounds

In this section, we first give a preliminary result, then structural properties for booking windows of 0, 1, and 2 days, and finally partial characterization of the optimal control of the general case. Some preliminary concepts and their definitions are given in the online supplement and all technical proofs are given in Appendix.

4.1. Preliminaries

Definition (Li & Yu, 2014): A set $V \subseteq \mathbb{Z}^n$ is called a polyhedron if there exist $\bar{\alpha}_i \in \mathbb{R}^n$ and $\beta_i \in \mathbb{R}$ such that $V = \{\bar{v} \in \mathbb{Z}^n | \bar{\alpha}_i^T \cdot \bar{v} \geq \beta_i, i = 1, 2, \dots, m\}$. Let (P1) be a special polyhedral form so that each n -dimensional vector $\bar{\alpha}_i$ has the form $\pm(0, \dots, 0, 1, \dots, 1, 0, \dots, 0)$; that is, the non-zero components of $\bar{\alpha}_i$ are either consecutive ones or consecutive negative ones.

The following preliminary result needed in our proofs is also of general interest.

Lemma 1. Let D be a polyhedron of integer vector $(\bar{x}, \bar{z}, \bar{y}) \in \mathbb{Z}^{n_1+m+n_2}$ satisfying (P1) and let $S \subseteq \mathbb{Z}^{n_1+n_2}$ be the domain of $(\bar{x}, \bar{y})$ and $A(\bar{x}, \bar{y}) \subseteq \mathbb{Z}^m$ that of $\bar{z}$ for any given $(\bar{x}, \bar{y})$. If $g(\bar{x}, \bar{z}, \bar{y})$ is multimodular (antimultimodular) in $(\bar{x}, \bar{z}, \bar{y})$ on D , then

$$f(\bar{x}, \bar{y}) = \min_{\bar{z}} \left(\max_{\bar{z}} \right) \left\{ g(\bar{x}, \bar{z}, \bar{y}) \mid \bar{z} \in A(\bar{x}, \bar{y}) \cap \left\{ \bar{z} \in \mathbb{Z}^m \mid \sum_{i=1}^m z_i = \gamma \right\} \right\}$$

is multimodular (antimultimodular) in $(\bar{x}, \bar{y})$ on S .

Lemma 1 states that, under an extra equality condition, the multimodularity (antimultimodularity) of a function is preserved by minimization (maximization) over a set of decision variables that lie in the middle of the list of ordered variables of the function. It is noteworthy that the order of variables is important for the proof of multimodularity (antimultimodularity) (Li & Yu, 2014). To our knowledge, in the current theory of discrete convex analysis, the preservation of multimodularity (antimultimodularity) under minimization (maximization) is only proved over the left side or right side of the list of ordered variables. No theories have hitherto supported the preservation of multimodularity (antimultimodularity) by optimization over the middle part of the list. Therefore, Lemma 1 fills the gap of the theory and is thus of general interest. Lemma 1 will be helpful in the proof of the main results of this paper.

In the following, we rewrite the optimality equation in an equivalent form that is more convenient for the theoretical analysis.

$$f_t(\bar{x}) = - \sum_{i=0}^{L-1} r_i x_i + E_{a_t} [B(\bar{x}; a_t)] \quad (3)$$

where

$$B(\bar{x}; a) = \max_{q_0 + \dots + q_L \leq a} Q_t(\bar{x}, \bar{q}) \quad (4)$$

$$Q_t(\bar{x}, \bar{q}) = g_t(x_0 + q_0) + F_{t+1}(q_1 + x_1, \dots, q_{L-1} + x_{L-1}, q_L) \quad (5)$$

$$g_t(y) = r_0 y - C_t(y) \quad (6)$$

$$F_{t+1}(y_1, \dots, y_L) = r_1 y_1 + \dots + r_L y_L + f_{t+1}(y_1, \dots, y_L) \quad (7)$$

Let s_{it} be the greatest value $y \geq 0$ that maximizes $r_i y - C_{t+i}(y)$.

4.2. Zero and one-day booking windows

The case of the null booking window, i.e., $L=0$, is trivial as the control is only over the current-day appointment and hence the optimal control consists of simply maximizing the stage reward. Given the concavity of $g_t(y)$, the optimal control is $q_0 = s_{0t} \wedge a_t$. Note that s_{0t} is the greatest $y \geq 0$ that maximizes $r_0 y - C_t(y)$.

We now consider the one-day booking window. The following theorem states that the optimal value function is concave.

Theorem 1. If $L=1$, then $f_t(x_0)$ is concave in x_0 .

The following theorem characterizes the admission control (q_0, q_1) and establishes the monotonicity and bounded sensitivity of (q_0, q_1) w.r.t. the state and the number of arrivals.

Theorem 2. If $L=1$, then there exists an optimal control such that $q_0 + q_1 = a_t \wedge ((s_{0t} - x_0)^+ + K_t)$, $q_1 = (a_t - q_0) \wedge K_t$, $-1 \leq \Delta_{x_0} q_0(x_0, a) \leq 0$, $0 \leq \Delta_{x_0} q_1(x_0, a) \leq 1$, and $0 \leq \Delta_a q_i(x_0, a) \leq 1$, $i = 0, 1$ where $K_t = \arg \max_{y \geq 0} F_{t+1}(y)$.

Theorem 2 allows the determination of the total number of admissions based on the critical queue lengths s_{0t} and K_t (respectively) the current day and the next. However, the partition between the two days is more subtle: a simple threshold rule of filling first the current day to s_{0t} is not optimal. From the bounded sensitivity, as the number of arrivals increases by 1, the additional admission if any is assigned to either the current day or the next without modifying the previous assignment. This provides insight into an alternative perspective of addressing the problem, i.e., obtaining the optimal assignment sequence for each arrival before knowing the actual arrivals, in contrast to determining the assignment after knowing all arrivals.

4.3. Two-day booking window

In this subsection, we consider the two-day booking window and present the theoretical results. The following theorem states that the objective function exhibits discrete concavity (antimultimodularity).

Theorem 3. If $L=2$, then $f_t(x_0, x_1)$ is antimultimodular in (x_0, x_1) .

We first demonstrate the assignment of n new arrivals to the next two days, i.e., $q_1 + q_2 = n$.

Theorem 4. Assume $L=2$. Let $\bar{X}(y, n) = (X_1(y, n), X_2(y, n))$ be a two-dimensional vector such that $\bar{X}(y, 0) = (y, 0)$ and $\bar{X}(y, n+1) = \bar{X}(y, n) + e_{\eta(\bar{X}(y, n))}$, where e_i is a vector of zero, except that the entry i is equal to 1 and $\eta(\bar{X}) = 1$ if $F_{t+1}(\bar{X} + e_1) \geq F_{t+1}(\bar{X} + e_2)$ and $\eta(\bar{X}) = 2$ otherwise. Then $\bar{X}(y, n) = (y + q_1, q_2)$ is an optimal solution of $\max_{q_1+q_2=n, (q_1, q_2) \geq 0} F_{t+1}(y + q_1, q_2)$, i.e., $\bar{X}(y, n)$ is the next booking state with y existing patients in day $t+1$ and n new patients for days $t+1$ and $t+2$. Further, $\bar{X}(y, n) = \bar{X}(0, y+n)$ if $X_1(0, y+n) \geq y$ and $\bar{X}(y, n) = (y, n)$ otherwise.

Theorem 4 shows that the optimal partition (q_1, q_2) of n new patients for the next two days and the resulting new booking state $\bar{X}(x_1, n)$ can be determined locally by optimally assigning each extra new patient without modifying the assignment of the previous new patients. Further, $\bar{X}(x_1, n)$ can be determined by starting from an empty booking state with $x'_1 = 0$ and by using the state-independent assignment sequence leading to $\bar{X}(0, x_1 + n)$. With this sequence, we are able to assign any new arrival to the optimal period based on the rule in Theorem 4.

Theorem 5. If $L=2$, then there exists an optimal control such that (i) $|\Delta_{x_i} q_j| \leq 1$, $\Delta_{x_i} q_i \leq 0$, $0 \leq \Delta_{x_0} (q_1 + q_2) \leq 1$, $-1 \leq \Delta_{x_1} (q_1 + q_2) \leq 0$, (ii) $\Delta_{x_0} (x_i + q_i) \geq 0$, $0 \leq \Delta_{x_0} (\sum_{i=0}^2 (x_i + q_i)) \leq 1$, (iii) $0 \leq \Delta_a q_i \leq 1$, $0 \leq \Delta_a (\sum_{i=0}^2 q_i) \leq 1$ and is non-increasing in a , and (iv) $0 \leq \Delta_{x_1} (x_i + q_i) \leq 1$, $\forall i \in \{0, 1\}$, $0 \leq \Delta_{x_1} (\sum_{i=j}^2 (x_i + q_i)) \leq 1$, $\forall j \in \{0, 1\}$.

Remark 1. The antimultimodularity of the optimal reward function implies its joint concavity and hence the global optimality of any local optimum. This property has been used in the derivation of the structural properties and will also be used to design heuristic policies for the general cases.

Remark 2. Whereas the unit increase of x_0 leads simply to a change of at most one unit for at most one $x_i + q_i$, the sensitivity w.r.t. x_1 is more subtle. A unit increase of x_1 leads to unit changes of at most two q_i , an eventual unit reduction of q_1 , an eventual unit increase of the preceding q_0 , and a unit increase for all cumulative booking $\sum_{i=j}^2 (x_i + q_i)$ covering period 1. $x_2 + q_2$ can either increase or decrease with a unit increase of x_1 .

Remark 3. The sensitivity w.r.t. a makes it possible to implement the optimal policy for dynamically-arriving patients during a day. Theorems 2 and 5 imply that the optimal control of the a -th patient can be made without changing the assignment of previous patients. It is thus possible to create a sequence of optimal assignment $\Delta_a \bar{q}$ for $a=0, 1, \dots$. The first patient is assigned to $\Delta_a \bar{q}|_{a=0}$, the second to $\Delta_a \bar{q}|_{a=1}$, and so on.

4.4. General booking windows

This subsection provides the bounds of the optimal control and partial characterization of the optimal control for the general case with booking windows of arbitrary lengths. Counter-examples of the antimultimodularity are also provided.

The following shows that the queue length after admission $x_i + q_i$ of day $t+i$ is at most $s_{it} \vee x_i$.

Theorem 6. If $t+L < N$, then $q_i \leq (s_{it} - x_i)^+$ with the convention $x_L = 0$.

Theorem 6 provides a bound of the state space. If the initial state is such that $x_{i0} \leq \bar{s}$, $\forall i$ with $\bar{s} = \max_{i,t} s_{it}$, then Theorem 2 implies that the booking state remains bounded by $\bar{s}$, $\forall i$, i.e., $x_{it} \leq \bar{s}$, $\forall i, t$.

The following theorem shows the relationship between the reward pattern and the optimal policy.

Theorem 7. Assume $t+L < N$. For all $i < j$ such that $q_j > 0$, either $r_i - r_j \leq \max_{k>0} (r_{i-k} - r_{j-k})$ or $r_i - \Delta_{C_{t+i}}(x_i + q_i) \leq r_j - \Delta_{C_{t+j}}(x_j + q_j - 1)$.

Remark 4. It is noteworthy that $r_i - r_j > \max_{k>0} (r_{i-k} - r_{j-k})$ implies that it is “better to move from j to i now than later” and $r_i - \Delta_{C_{t+i}}(x_i + q_i) > r_j - \Delta_{C_{t+j}}(x_j + q_j - 1)$ implies a “higher marginal reward for adding the last patient to i than to j ”. The condition $r_i - \Delta_{C_{t+i}}(x_i + q_i) \leq r_j - \Delta_{C_{t+j}}(x_j + q_j - 1)$, $\forall i, j : q_j > 0$ is the optimality condition of the myopic optimum without considering future arrivals. It tends to uniformize the marginal reward. By imposing this condition only to $i < j$, Theorem 7 shows that the global optimal policy schedules the new patients earlier than the myopic optimum. To reserve capacity for future patients, the global optimal policy accepts fewer patients than does the myopic optimum which accepts $a_t \wedge \sum_{i=0}^L (s_{it} - x_i)^+$ patients.

Remark 5. In the case of the stationary reward $r_i = r$, $\forall i$, conditions of Theorem 7 become $\Delta_{C_{t+i}}(x_i + q_i) \geq \Delta_{C_{t+j}}(x_j + q_j - 1)$, $\forall i < j : q_j > 0$. As a result, if the booking state has the property $\Delta_{C_{t+i}}(x_i) \geq \Delta_{C_{t+j}}(x_j - 1)$, $\forall i < j : x_j > 0$ that we call marginal cost decreasing, then $\Delta_{C_{t+i}}(x_i + q_i) \geq \Delta_{C_{t+j}}(x_j + q_j - 1)$, $\forall i < j : x_j + q_j > 0$ and the new booking state remains marginal cost decreasing. If otherwise, $\Delta_{C_{t+i}}(x_i + q_i) \geq \Delta_{C_{t+L}}(q_L - 1)$, $\forall i < L : q_L > 0$ and the booking state becomes marginal cost decreasing after L days. If the capacity usage cost is also stationary, i.e., $C_t(y) = C(y)$, then $\Delta_{C_{t+i}}(x_i) > \Delta_{C_{t+j}}(x_j - 1)$ implies $x_i \geq x_j$ and marginal cost decreasing implies booking state decreasing, i.e., $x_0 \geq \dots \geq x_{L-1} \geq x_L = 0$. As a result, if the booking state is decreasing, then the new booking state remains decreasing. Otherwise, the booking state becomes decreasing after L days.

The following theorem establishes the relationship between the optimal total acceptance and myopic policy under the condition of the stationary reward.

Theorem 8. If the reward is stationary $r_i = r$ and $t+L < N$, then there exists an optimal control such that $q_0 + \dots + q_L = a_t \wedge Z(\bar{x})$ where $Z(\bar{x}) = \sum_{i=0}^L (s_{0,t+i} - x_i)^+$.

From [Theorem 8](#), whereas patients may be rejected now to reserve capacity for higher-reward future patients in case of non-stationary reward, the issue of whether to reject current demand to save capacity for future arrivals is irrelevant and the determination of the total number of patients to accept is trivial if the rewards are stationary. A patient is accepted if it is myopically beneficial to do so, i.e., if the marginal reward is positive. We do not need to proactively reject some patients in anticipation of future arrivals. Of course, the distribution of the accepted patients in the booking window is still nontrivial and depends on future arrivals and is marginal cost decreasing.

Remark 6. As we are interested in long-run optimum, the border case of $t + L > N$ and $t < N$ is of marginal interest. In this case, all patients will be accepted, as their capacity usage cost is not considered. Nevertheless, such border case can be avoided by extending the time horizon to cover the booking window of the last patients without additional arrivals.

Counter-examples of the antimultimodularity: For $L \geq 3$, the optimal reward function $f_t(\bar{x})$ is not antimultimodular. Extensive numerical experiments for booking windows of 3 and 4 days have been conducted to check the necessary conditions of antimultimodularity in (Li & Yu, 2014; Lemma 1). Necessary conditions are shown to fail in the online supplement for $L = 3$: $a_t \sim \text{Poi}(3)$, $r_i = r \in \{0.1, 0.99, 1.0, 1.2\}$, $C_t(y) = E_\varepsilon[(y + \varepsilon - 5)^+]$ with $\varepsilon \sim \text{Poi}(1)$. As a result, the optimal reward function is in general not antimultimodular even for stationary reward and stationary arrivals.

5. Heuristic policies

This section proposes two heuristic policies founded upon analytical results from [Section 4](#) and several other simple heuristic policies for comparison.

Assignment-Sequence Policy (ASP): The ASP relies on a single assignment sequence $\bar{S} = [S_k]_{k=1}^\infty$, where $S_k \in \{-1, 0, 1, \dots, L\}$. For any state $\bar{x}$ and a new arrivals, a new arrival is either rejected if $S_k = -1$ or is admitted to day i if $S_k = i$ and $\sum_{j=1}^k 1(S_j = i) > x_i$. The assignment starts from $k=1$ and is repeated until decisions for all a new arrivals are done. For example, with $L=1$, $x_0=2$ and $\bar{S} = [0, 1, 0, -1, 1, 0, 1, -1, -1, \dots]$, $q_0=0$ and $q_1=1$ for two arrivals $a=2$ since $S_1=0$ and $\sum_{j=1}^k 1(S_j = 0) = 1 \leq x_0$ for $k=1$, $S_2=1$ and $\sum_{j=1}^k 1(S_j = 1) = 1 \geq x_1$ leading to **assignment to day $t+1$** for $k=2$, $S_4=-1$ leading to **rejection** for $k=4$. Similarly, $q_0=1$ and $q_1=2$ for $a=4$.

The ASP is inspired by the special case of $L=2$ for which the optimal admission can be determined locally one new arrival at a time and can be made through the optimal assignment sequence at the empty booking state $\bar{x} = \bar{0}$.

$\bar{S} = [S_k]_{k=1}^\infty$ is the assignment sequence for state $\bar{x} = \bar{0}$ and the assignment S_k of the k -th new arrival is determined locally without modifying the previous assignments $S_1, \dots, S_{k-1}$. Each alternative of $S_k \in \{-1, 0, \dots, L\}$ is evaluated by simulation over a look-ahead window of M periods with k arrivals of the current period t assigned according to $\bar{S}$ and randomly generated future arrivals a_{t+i} admitted according to some given heuristic policy that performs well (MTP or MP in our numerical experiments). The final S_k is the one that maximizes the total reward estimated through sample average approximation for a given set Ω of scenarios. In our numerical experiments, we set $M = 10L$ and $|\Omega| = 1000$. By construction, the sequence determined locally is such that $S_k = -1$ implies $S_{k+i} = -1$. The computational cost of establishing the sequence $\bar{S}$ offline is $K|\Omega|$ simulation runs over M periods where K is the smallest k such that $S_k = -1$ and $K \leq \sum_{i=0}^L s_{it}$ by [Theorem 6](#).

Multiple-Threshold Policy (MTP): The MTP is defined by $(L+1)$ booking thresholds $n_0, \dots, n_L$ and assigns new arrivals as early as

possible with no admission for days $x_i \geq n_i$ and booking after new assignments not exceeding n_i for the other days. It is motivated by [Theorem 6](#) which demonstrates that the admission for day i of the booking window is upper-bounded by $q_i \leq (s_{it} - x_i)^+$. The MTP is evaluated by simulation and is set by exhaustive search for small-sized cases and local search for the others within the bounds of [Theorem 6](#).

Single-Threshold Policy (STP): The STP is an MTP with equal thresholds, i.e., $n_0 = \dots = n_L = n$. It is commonly used in practice since it is easier both to determine and to understand.

Myopic Policy (MP): The MP makes the admission decision without considering future arrivals, i.e., $\max_{q_0+\dots+q_L=a} \sum_{i=0}^L (r_i q_i - C_{t+i}(x_i + q_i))$. By concavity, new arrivals are individually assigned to the day of the greatest marginal reward $\Delta_{q_i}(r_i q_i - C_{t+i}(x_i + q_i))$ until it becomes negative.

First-Come-First-Served Policy (FCFS): This rule applies to the case with a regular capacity. A given period is said to be available if the number of patients assigned to it is less than the capacity. For this commonly-used policy, patients are assigned to the first period with an available capacity until all the periods become unavailable. In the case with emergency arrivals, in contrast to the other policies, the FCFS policy does not reserve any capacity for emergency patients.

6. Numerical experiments

This section is dedicated to the numerical comparison and sensitivity analysis of heuristic policies of [Section 5](#). The comparison is done w.r.t. the optimal policy for small-sized cases and the best heuristic policy for cases of realistic sizes.

6.1. Experimental setting

In the numerical experiments, we consider the scheduling of outpatient imaging examination through diagnostic equipment shared with emergency patients who are served on the day of their arrival. The daily capacity is divided into multiple equal-length time slots and each patient is served by one slot. The arrivals of outpatients and emergency patients are mutually independent and assumed to follow Poisson distributions, respectively with the means of λ and of μ . The stage reward and cost structure are defined as $\sum_{i=0}^L r_i q_i - E_e[(x_0 + q_0 + e - H)^+]$, where e indicates a random number of emergency patients and H indicates the regular capacity, e.g. time slots or unit of machines. The first term indicates the reward for accepting regular patients. The second indicates the cost of using extra time slots and is increasing convex. In practice, the regular capacity for 1 unit of machine per day is 8 h. We set $H=32$, from which the implication is twofold: the time slot for an examination is equal either to 15 min given one unit of machine, or to 30 min given two units of machines. Overcapacity cost is used as the capacity usage cost. Unit overcapacity cost and unit rewards are scaled such that the former is equal to 1. The goal is to maximize the long-run average net reward as evaluated by a long simulation of 10^6 days plus 10^4 days of warm-up under common random arrivals for all policies. The estimation is precise enough for statistical validity of the numerical observations of this paper. For example, for Base Case 2 defined below, the 99.9%-confidence interval of the average reward of ASP policy for 10 independent simulations is 21.086615 ± 0.002525 .

Base cases for the small-sized cases and the large-scale cases are defined as follows.

- 1) Base Case 1 (Small-sized case): $L=2$, $H=32$, $\lambda=32$, $\mu=5$, $r_0=0.8$, $r_1=0.75$, $r_2=0.7$;
- 2) Base Case 2 (Large-scale case): $L=14$, $H=32$, $\lambda=32$, $\mu=5$, $r_i=0.8 - i/100$, $i=0, 1, \dots, L$;

Table 1
Percentage optimality gaps for small-sized instances.

Parameter		Reward_ OPT	Gap (%)				
			ASP	MTP	STP	MP	FCFS
Reward pattern	RP-S	21.21	0.00	0.00	0.00	0.19	2.07
	RP-D	20.90	0.00	0.00	3.44	12.01	8.33
	RP-C	17.84	0.00	0.00	0.35	0.13	4.94
Regular arrival rate λ	24	18.82	0.01	0.09	0.10	4.86	2.41
	32	20.90	0.00	0.00	3.44	12.01	8.33
	40	21.19	0.00	0.00	9.67	13.32	16.90
Emergency arrival rate μ	5	20.90	0.00	0.00	3.44	12.01	8.33
	10	16.75	0.00	0.00	7.46	13.98	16.03
	15	12.56	0.00	0.00	15.63	15.49	28.04
Length of booking window L	1	20.90	0.00	0.01	3.44	5.62	4.51
	2	20.90	0.00	0.00	3.44	12.01	8.33

The following numerical experiments are conducted:

Sensitivity *w.r.t.* the outpatients or emergency arrival rate: Base Cases 1 and 2 with $\lambda = \{24, 32, 40\}$ or $\mu = \{5, 10, 15\}$;

Sensitivity *w.r.t.* the reward pattern (RP): (A) Base Cases 1 with: i) RP-S: stationary reward pattern with $r_0 = 0.8, r_1 = 0.8, r_2 = 0.8$; ii) RP-D: decreasing reward pattern with $r_0 = 0.8, r_1 = 0.75, r_2 = 0.7$; and iii) RP-C: cyclic reward pattern with $\vec{r} = (0.8, 0.5, 0.8, 0.5, 0.8)$, which means that patients value their booking date *w.r.t.* the exact day of week rather than how far it is from the current day. (B) Base Case 2 with: i) RP-S: stationary reward pattern with $r_i = 0.8, i = 0, 1, \dots, L$; ii) RP-D: decreasing reward pattern with $r_i = 0.8 - i/100, i = 0, 1, \dots, L$; and iii) RP-C: cyclic reward pattern with the reward cycle $\vec{r} = (0.8, 0.5, 0.8, 0.5, 0.8)$. Note that the cyclic reward pattern is related to the exact day instead of the day in the moving window and is chosen to consider the preference of the days of the week, e.g. Monday, Wednesday, and Friday.

Sensitivity *w.r.t.* the length of the booking window (L): Base Case 1 with $L=\{1,2\}$ and the reward pattern is (0.80,0.75) if $L=1$, or Base Case 2 with $L=\{7,14,21,28\}$. For the small-sized case, we consider only $L=1$ or 2 days because, when $L>2$, it becomes difficult to determine the optimal policy within an acceptable computation time.

6.2. Comparison of performances

This subsection first compares the long-term average reward of the heuristic policies (ASP, MTP, STP, MP, and FCFS). The degree of optimality is measured by a gap defined as $(1 - \frac{\text{Reward of heuristic policy}}{\text{Reward of benchmarked policy}}) \times 100\%$, which denotes percentage deviations from the optimum for small-sized instances and percentage deviations from the ASP for large-scale instances. Later in this subsection, we compare other performance measures including the average admission (for regular patients), average occupancy (including regular and emergency patients), average overcapacity, and average idle time (or equivalently, underutilization).

Table 1 shows the results of different heuristic policies for the small-sized problem. The ranking of the heuristic policies in most cases is as follows: ASP (best), MTP, STP, MP, and FCFS (worst). The performance of the ASP is the closest to the optimal policy with the largest gap of 0.01%, followed by the MTP with the largest gap of 0.09%. The STP and MP perform poorly with the largest gaps of 15.63% and of 15.49% respectively, since the former uses a single threshold for the acceptance of all periods and the latter considers only the current state and reward. The worst policy, the FCFS, is outperformed by all the other policies in most cases, with its optimality gap between [2.07%, 28.04%]. Under the decreasing reward pattern (RP-D), it is noteworthy that the gaps of the STP, MP, and FCFS increase with the regular or emergency arrival rate, implying

Table 2
Percentage gaps from ASP for large-scale instances.

Parameter		Reward_ ASP	Gap (%)			
			MTP	STP	MP	FCFS
Reward pattern	RP-S	21.21	0.00	0.00	0.00	2.00
	RP-D	21.09	0.03	3.09	18.23	12.82
	RP-C	17.84	0.01	0.35	0.00	4.99
Regular arrival rate λ	24	19.01	0.08	0.05	7.40	3.33
	32	21.09	0.03	3.09	18.23	12.82
	40	21.19	0.02	9.67	18.71	23.01
Emergency arrival rate μ	5	21.09	0.03	3.09	18.23	12.82
	10	16.76	0.00	7.41	19.86	20.99
	15	12.56	0.29	15.76	21.62	34.64
Length of booking window L	7	21.09	0.01	3.09	8.94	6.94
	14	21.09	0.03	3.09	18.23	12.82
	21	21.08	0.34	3.36	27.49	18.42
	28	21.02	0.52	3.12	36.48	23.23

Table 3
Percentage optimality gaps under cyclic reward pattern for small-sized instances.

Parameter		Reward_ OPT	Gap (%)				
			ASP	MTP	STP	MP	FCFS
λ	8	6.35	0.00	0.04	15.00	0.00	15.00
	16	12.43	0.02	0.76	12.43	0.82	12.54
	24	16.46	0.08	0.92	1.54	0.52	5.86
	32	17.84	0.00	0.00	0.35	0.13	4.94
	40	17.84	0.00	0.05	0.35	0.00	5.00
	48	17.84	0.00	0.01	0.35	0.00	5.00

the progressively significant advantages of the ASP and MTP given rising workloads. Under the stationary reward pattern (RP-S), the performances of all heuristic policies except the FCFS are the same or nearly the same as that of the optimal policy. This observation repeats itself in more experiments with different parameters.

Table 2 compares the percentage deviation from the ASP for the large-scale instances. The ASP outperforms all its heuristic counterparts for all instances. The MTP performs slightly worse than the ASP with a percentage gap of [0.00%, 0.52%]. These results show that both policies yield excellent performances consistently in both small-sized and large-scale cases. As shown in Table 1, under the decreasing reward pattern (RP-D), the gaps of the STP, MP, and FCFS increase with the regular or emergency arrival rate; under the stationary reward pattern (RP-S), the performances of all policies except FCFS are the same.

Through a joint analysis of Tables 1 and 2, we elucidate the effect of a longer booking window, i.e., larger L , on the performances of the heuristic policies. When the booking window increases from 1 day to 28 days, the observations are: the optimal reward and the reward of the ASP are steady; and the gaps of the MP and FCFS from the optimal policy or the ASP continuously increase. The implication is that, given a prolonged booking window, the performances of both the MP and FCFS deteriorate.

We also conduct more experiments under the cyclic reward pattern. Since the problem becomes more complex, the length of the look-ahead periods for the ASP is expanded to $M=50L$ in this case. We focus on the sensitivity *w.r.t.* the regular arrival rate. Table 3 outlines the percentage optimality gaps of all the heuristic policies and Table 4 outlines the percentage gaps of the policies against the ASP. Numerous observations are of note. Firstly, by inspection of Table 3, when the regular arrival rate falls below 24, the optimality gaps of the STP and FCFS rise above 10% whereas those of the ASP and MTP remain smaller than 1%. This means that, given the low outpatient arrival rate under the cyclic reward pattern, the ASP and MTP perform favorably. When the workloads are low, poor advance-scheduling decisions will result in both the underutilization of the resources and overtime. There-

Table 4
Percentage gaps against ASP under cyclic reward pattern for large-scale instances.

Parameter		Reward_ ASP	Gap (%)			
			MTP	STP	MP	FCFS
λ	8	6.34	0.23	15.00	0.00	15.00
	16	12.46	2.08	12.73	0.12	12.81
	24	16.47	0.87	1.27	0.00	5.90
	32	17.84	0.01	0.35	0.00	4.99
	40	17.84	0.00	0.35	0.00	5.00
	48	17.84	0.00	0.35	0.00	5.00

fore, advance scheduling becomes more important in the context of low workloads. Secondly, likewise by inspection of Table 3, the optimality gaps of the STP and FCFS decrease with escalating regular arrival rates under the cyclic reward pattern, in contrast to that under the decreasing reward pattern. Thirdly, Table 4 reveals the progressively significant advantages of the ASP and MTP given the decreasing outpatient arrival rate. Lastly, and intriguingly, the MP emerges as an excellent policy under the cyclic reward pattern.

We next thoroughly compare the performances of the policies based on detailed results given in Tables C1–C5 of the online supplement. The performance indicators of the policies for small-sized problems are tabulated in Tables C-1 to C-4; and those for large-scale problems are in Table C-5. The average daily admission, occupancy, and overcapacity of the optimal policy, ASP, and MTP are the lowest among all the policies for most instances. Besides, the idle time of these three policies are less than 0.5 time slots in most cases. The lowest average overcapacity is associated with the optimal policy for most instances, whereas the reverse is true for the FCFS: the inferior performance of FCFS is attributed to its lack of consideration for reserving capacity for emergency patients. The above observations collectively demonstrate that the performance indicators are well-balanced for the optimal policy and the near-optimal ones (the ASP and MTP). It should be highlighted that, as shown in Table C-5, all the performance indicators do not change with the longer booking window.

We now consider the limiting case of infinite arrivals, i.e. $\lambda = \infty$. For reward r_i that depends only on the position i in the moving time window, it is optimal to assign arrivals of period t only to the period $t + i^*$ with the highest reward r_i and the optimal average reward is then $\max_q r_{i^*} q - E_e[(q + e - H)^+]$ equal to 21.21 for the cases of Tables 1–2. For the cyclic reward r_{t+i} that depends only on the day of the service $t + i$, it is optimal to admit only the same day and, with K being the cycle length, the optimal daily net reward becomes $\frac{1}{K} \sum_{t=1}^K \max_q r_t q - E_e[(q + e - H)^+]$ equal to 17.84 for the cases of Tables 3–4.

Finally, we offer insights into the effects of the parameters on the performance indicators. An elevated outpatient arrival rate translates into the higher average admission, average occupancy, and average overcapacity but a lower idle time for each policy. An elevated emergency arrival rate translates into the lower average admission for each policy (except for FCFS) but higher average occupancy and overcapacity for each policy.

7. Conclusions

This paper addresses the advance-scheduling problem of outpatients within a booking window considering future random arrivals. Patients can either be scheduled to a date within the booking window or be rejected. A finite-horizon MDP model is proposed to maximize the total expected reward for accepting regular patients minus the capacity usage cost. The structural properties of the optimal objective function and the optimal control policy are established. For the optimal objective function, its antimultimodularity is proved when the length of the booking window is equal

to 1 or 2 periods, with the implication that the locally-optimal decision is also the globally-optimal one. For the optimal policy, its monotonicity and bounded sensitivity are established. Further, several bounds on the optimal decision are developed. Inspired by the theoretical results, we propose two main heuristic policies, including the assignment-sequence policy and multiple-threshold policy. Through simulation, both policies are compared with the optimal policy for the small-sized cases and compared with several simple heuristic policies for the large-scale cases. Numerical experiments show that the assignment-sequence policy and multiple-threshold policy not only are easy to implement but also present a good approximation of the optimal policy for the small-sized cases and perform the best against the heuristic policies for the large-scale cases. Further, these two policies are realistic because they are able to schedule each arrival without knowing the number of all arrivals in advance. Lastly, filling the gap of the theory of discrete convex analysis, we prove that, multimodularity (antimultimodularity) is preserved under minimization (maximization) over the middle part of the list of ordered variables with an additional equality constraint.

This paper provides insights into scheduling outpatients to periods within a booking window. Firstly, the locally-optimal decision is not only likewise the globally-optimal one for a small special case but also a good approximation of the globally-optimal one in special cases; accordingly, the locally-optimal decision can be expected to perform well for other cases. Secondly, the number of outpatients scheduled on each day should be less than a specific bound that can be easily determined by Theorem 6. Thirdly, from Theorem 7, the workload of patients after the decision should be non-increasing from the current period to the last period in the booking window, provided that the reward pattern and the capacity usage cost are both stationary. More insights are gained from the numerical experiments. (i) For general reward patterns, it is beneficial to consider multiple booking thresholds for different periods within the booking window rather than a single threshold. (ii) The assignment sequence for each arrival is important for non-stationary reward patterns and the advantage is significant especially when the arrival rate is expected to be low. (iii) When considering randomly-arriving emergency patients, FCFS is a somewhat underperforming policy; the assignment-sequence policy or multiple-threshold policy can be used for improving the performance.

This research can be extended in several directions. One direct extension is to consider multiple types of patients and multiple waiting-time targets for them instead of a single booking window. Another one is to consider multiple resource requirements. A third direction is to incorporate patient behaviors such as cancellations and no-shows.

Acknowledgments

This work was supported by the National Natural Science Foundation of China [grant numbers 72031007, 71972129, 71671111].

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2020.11.030.

Appendix

Proof of Lemma 1. We only consider the multimodularity and the proof for the antimultimodularity is similar. Firstly, by (Hajek, 1985, Proposition 2.2), the multimodularity of $g(\vec{x}, \vec{z}, \vec{y})$ implies the submodularity of $h(u, \vec{x}, \vec{z}, \vec{y}) = g(x_1 - u, x_2 - x_1, \dots,$

$x_{n_1} - x_{n_1-1}, z_1 - z_{n_1}, z_2 - z_1, \dots, z_m - z_{m-1}, y_1 - z_m, y_2 - y_1, \dots, y_{n_2} - y_{n_2-1}$ in $(u, \bar{x}, \bar{z}, \bar{y})$ with $u \in \mathbb{R}$.

$f(\bar{x}, \bar{y})$ is multimodular in $(\bar{x}, \bar{y})$, if $\psi(u, \bar{x}, \bar{y})$ defined as follows is submodular:

$$\begin{aligned} \psi(u, \bar{x}, \bar{y}) \\ = f(x_1 - u, x_2 - x_1, \dots, x_{n_1} - x_{n_1-1}, y_1 - x_{n_1}, y_2 - y_1, \dots, y_{n_2} - y_{n_2-1}) \end{aligned}$$

By definition, $\psi(u, \bar{x}, \bar{y})$ can be written as follows:

$$\psi(u, \bar{x}, \bar{y}) = \min_{(\bar{z}, \bar{y}) \in \Omega(u, \bar{x}, \bar{y}; \gamma)} g \left(\begin{matrix} x_1 - u, x_2 - x_1, \dots, x_{n_1} - x_{n_1-1}, \\ z_1 - x_{n_1}, z_2 - z_1, \dots, z_m - z_{m-1}, \\ \bar{y}_1 - z_m, \bar{y}_2 - \bar{y}_1, \dots, \bar{y}_{n_2} - \bar{y}_{n_2-1} \end{matrix} \right)$$

where $\Omega(u, \bar{x}, \bar{y}; \gamma)$ is the section of $\Omega(\gamma)$ at $(u, \bar{x}, \bar{y})$ and

$$\Omega(\gamma) = \left\{ \begin{matrix} (u, \bar{x}, \bar{z}, \bar{y}, \bar{y}) \in \mathbb{Z}^{n_1+2n_2+m+1} \\ (x_1 - u, x_2 - x_1, \dots, x_{n_1} - x_{n_1-1}, y_1 - x_{n_1}, y_2 - y_1, \dots, y_{n_2} - y_{n_2-1}) \in S \\ (z_1 - x_{n_1}, z_2 - z_1, \dots, z_m - z_{m-1}) \in A(x_1 - u, \dots, x_{n_1} - x_{n_1-1}, y_1 - x_{n_1}, \dots, y_{n_2} - y_{n_2-1}) \\ z_m - x_{n_1} = \gamma \\ \bar{y}_i - y_i = \gamma, i = 1, \dots, n_2 \end{matrix} \right\}$$

$\Omega(\gamma)$ is obviously a lattice. The preservation of submodularity by minimization implies the submodularity of $\psi(u, \bar{x}, \bar{y})$ (Topkis, 1998, Corollary 2.6.2) and hence the multimodularity of $f(\bar{x}, \bar{y})$.

Proof of Theorem 1. Since $f_N(x_0) = 0, \forall x_0$ is concave, it is enough to show that the concavity of $f_{t+1}(x_0)$ implies that of $f_t(x_0)$. Assume $f_{t+1}(x_0)$ be concave. Since

$$B(x_0; a) = \max_{q_0+q_1 \leq a} g_t(x_0 + q_0) + r_1 q_1 + f_{t+1}(q_1) \quad (8)$$

By concavity of $g_t(y)$ and $f_{t+1}(y)$, $g_t(x_0 + q_0)$ is antimultimodular in (x_0, q_0) and hence in (x_0, q_0, q_1) and $r_1 q_1 + f_{t+1}(q_1)$ is also antimultimodular in q_1 and in (x_0, q_0, q_1) by (Li & Yu, 2014, Lemma 2), leading to the antimultimodularity of $g_t(x_0 + q_0) + r_1 q_1 + f_{t+1}(q_1)$. By (Li & Yu, 2014; Theorem 1), $B(x_0; a)$ is antimultimodular hence concave in x_0 and hence so is $f_t(x_0)$.

Proof of Theorem 2. First, we select q_0 as the greatest one among all optimal control policies and $q_1 = (a_t - q_0) \wedge K_t$ thanks to the concavity of $F_{t+1}(y)$. By concavity of $g_t(y)$, $q_0 + q_1 = a_t \wedge ((s_{0t} - x_0)^+ + K_t)$. As a result, the optimality equation implies

$$B(x_0, a) = \max_{q_0 \leq a} g_t(x_0 + q_0) + H_{t+1}(a - q_0) \quad (9)$$

where $H_{t+1}(y) = F_{t+1}(y), \forall y \leq K_t$ and $H_{t+1}(y) = F_{t+1}(K_t), \forall y > K_t$. By definition, $H_{t+1}(y)$ is concave and hence $g_t(x_0 + q_0) + H_{t+1}(a - q_0)$ is antimultimodular in (x_0, q_0) . By (Li & Yu, 2014; Theorem 1), $-1 \leq \Delta_{x_0} q_0(x_0, a) \leq 0$. From $q_1 = (a_t - q_0) \wedge K_t, 0 \leq \Delta_{x_0} q_1(x_0, a) \leq 1$.

In order to analyze the effect of a , put (9) in the following form:

$$B(x_0; a) = \max_{-a+q_0 \leq 0} g_t(x_0 + q_0) + H_{t+1}(-(-a + q_0)) \quad (10)$$

By concavity of $H_{t+1}(y)$, $H_{t+1}(-y)$ is concave and $H_{t+1}(-(-a + q_0))$ antimultimodular in $(-a, q_0)$ and so is $g_t(x_0 + q_0) + H_{t+1}(-(-a + q_0))$. By (Li & Yu, 2014; Theorem 1), $0 \leq \Delta_a q_0(x_0, a) \leq 1$. Since $q_1 = (a_t - q_0) \wedge K_t, 0 \leq \Delta_a q_1(x_0, a) \leq 1$.

Proof of Theorem 3. Firstly, $f_N(\bar{x}) = 0, \forall \bar{x}$ is obviously antimultimodular. The theorem is established by induction by proving that the antimultimodularity of $f_{t+1}(x_0, x_1)$ implies that of $f_t(x_0, x_1)$. Assume $f_{t+1}(x_0, x_1)$ is antimultimodular. Introduce a slack variable q' such that $q_0 + q' + q_1 + q_2 = a$.

Rewrite (5) as follows: $Q_t(x_0, q_0, q', q_2, q_1, x_1) = g_t(x_0 + q_0) + F_{t+1}(q_1 + x_1, q_2)$. Since $g_t(y)$ is concave in y , $g_t(x_0 + q_0)$ is antimultimodular in (x_0, q_0) by (Li & Yu, 2014, Lemma 2v) and hence antimultimodular in $(x_0, q_0, q', q_2, q_1, x_1)$. Since $f_{t+1}(y_1, y_2)$ is antimultimodular in (y_1, y_2) , $F_{t+1}(y_1, y_2)$ is also so and hence antimultimodular in (x_1, q_1, q_2) and in (q_2, q_1, x_1) by (Li & Yu, 2014, Lemma

2vi) implying the antimultimodularity in $(x_0, q_0, q', q_2, q_1, x_1)$ of $F_{t+1}(q_1 + x_1, q_2)$. As a result, $Q_t(x_0, q_0, q', q_2, q_1, x_1)$ is antimultimodular in $(x_0, q_0, q', q_2, q_1, x_1)$ by (Li & Yu, 2014, Lemma 2iii).

By Lemma 1,

$$B(x_0, x_1; a) = \max_{q_0+q'+q_1+q_2=a, q_i \geq 0, q' \geq 0} Q_t(x_0, q_0, q', q_2, q_1, x_1) \quad (11)$$

is antimultimodular in (x_0, x_1) . Finally, the preservation of antimultimodularity by expectation (Li & Yu, 2014, Lemma 2iv) implies the antimultimodularity of $f_t(x_0, x_1)$ in (x_0, x_1) .

Proof of Theorem 4. We prove the first part by induction. It is true for $n=0$ and assume it true for $\bar{X}(y, n)$. Let $i_n = \eta(\bar{X}(y, n))$ and hence $\bar{X}(y, n+1) = \bar{X}(y, n) + e_{i_n}$.

Let $U(x_1, x_2) = F_{t+1}(y + x_1, x_2) - M|x_1 + x_2 - n - 1|$ with $M > \sup_{x_1+x_2 \leq n+1} |F_{t+1}(y + x_1, x_2)|$. By definition, maximize $U(x_1, x_2)$ in

$(x_1, x_2) \in \mathbb{N}^2$ is equivalent to maximize $F_{t+1}(y + q_1, q_2)$ subject to $q_1 + q_2 = n + 1$. Further, $U(x_1, x_2)$ is also antimultimodular from Theorem 3. By (Altman, Gaujal, & Hordijk, 2000, Corollary 2.1), some linear extension of $U(x_1, x_2)$ is jointly concave and $\bar{z} = \bar{X}(y, n) + e_{i_n}$ is a global maximum, i.e., $\bar{X}(y, n+1) = \bar{z}$ if it is a local maximum, i.e., $U(\bar{z}) \geq U(\bar{z} + \delta)$ for all $\bar{z} + \delta \in \mathbb{N}^2$ and $\delta \in \{\pm e_1, \pm e_2, \pm(1, 1), \pm(1, -1)\}$. By definition, $U(\bar{z}) \geq U(\bar{z} + \delta), \forall \delta \in \{\pm e_1, \pm e_2, \pm(1, 1)\}$ and $U(\bar{z}) \geq U(\bar{z} - e_{i_n} + e_j), j \neq i_n$. We only need to check $U(\bar{z}) \geq U(\bar{z} + e_{i_n} - e_j)$ which is true as

$$U(\bar{z} - e_{i_n}) + U(\bar{z} + e_{i_n} - e_j) \leq U(\bar{z}) + U(\bar{z} - e_j)$$

$$U(\bar{z} - e_{i_n}) = U(\bar{X}(y, n)) \geq U(\bar{X}(y, n) + e_{i_n} - e_j) = U(\bar{z} - e_j)$$

where the first inequality is from the antimultimodularity and the second from the optimality of $\bar{X}(y, n)$.

Consider now the property

(P) $\bar{X}(y, n) = \bar{X}(0, y + n)$ if $X_1(0, y + n) \geq y$ and $\bar{X}(y, n) = (y, n)$ otherwise.

(P) clearly holds for $n=0$. Assume that (P) holds for n . If $X_1(0, y + n + 1) \geq y$, the unconstrained optimum $\bar{X}(0, y + n + 1)$ is feasible for the constrained problem $\bar{X}(y, n + 1)$ and hence is constrained optimum, i.e., $\bar{X}(y, n + 1) = \bar{X}(0, y + n + 1)$. Assume $X_1(0, y + n + 1) < y$. By the first part of the theorem, $X_1(0, y + n) \leq X_1(0, y + n + 1) < y$ and by induction assumption, $\bar{X}(y, n) = (y, n)$.

Let $\bar{X}^\circ = \bar{X}(0, n + y + 1) - e_2$. By definition of $\bar{X}(0, n + y + 1)$ and the first part of the theorem, the optimal control $\eta(\bar{X}^\circ) = 2$ and

$$F_{t+1}(\bar{X}^\circ + e_1) - F_{t+1}(\bar{X}^\circ + e_2) \leq 0, X_1^\circ < X_1(y, n) = y$$

By joint concavity of the linear extension of $F_{t+1}(\bar{X})$,

$$\begin{aligned} 0 &\geq F_{t+1}(\bar{X}^\circ + e_1) - F_{t+1}(\bar{X}^\circ + e_2) \\ &\geq F_{t+1}(\bar{X}^\circ + (y - X_1^\circ)(e_1 - e_2) + e_1) \\ &\quad - F_{t+1}(\bar{X}^\circ + (y - X_1^\circ)(e_1 - e_2) + e_2) \\ &= F_{t+1}(\bar{X}(y, n) + e_1) - F_{t+1}(\bar{X}(y, n) + e_2) \end{aligned}$$

which implies $\eta(\bar{X}(y, n)) = 2$ and hence $\bar{X}(y, n + 1) = \bar{X}(y, n) + e_2 = (y, n + 1)$. To summarize, (P) holds for $n + 1$ and is hence true by induction.

Proof of Theorem 5. Firstly, we transform $q_0 + q_1 + q_2 \leq a$ into $q'_0 + q_1 + q_2 = a$. By concavity of $g_t(y)$ and the definition of s_{0t} , the optimal control is such that $q_0 = q'_0 \wedge (s_{0t} - x_0)^+$ with $q'_0 =$

$a - q_1 - q_2$. As a result, the optimal control can be equivalently determined as follows

$$\max_{q_0+q_1+q_2=a} h_t(x_0 + q'_0) + F_{t+1}(q_1 + x_1, q_2) \quad (12)$$

where $h_t(y) = g_t(y \wedge s_{0t})$ is concave. Note that the optimal reward of the above is not equal to $f_t(\bar{x})$.

We then show the pointwise-concavity and supermodularity of $H_{t+1}(y, z) = \max_{z+q_1+q_2=0} F_{t+1}(q_1 + y, q_2)$. $F_{t+1}(q_1 + y, q_2)$ is antimultimodular in (q_1, q_2) and in (z, q_1, q_2) leading to the antimultimodularity and hence concavity of $H_{t+1}(y, z)$ in z by (Li & Yu, 2014; Theorem 1). Similarly $F_{t+1}(q_1 + y, q_2)$ is antimultimodular in (y, q_1, q_2) leading to the antimultimodularity and concavity of $H_{t+1}(y, z)$ in y . Further $H_{t+1}(y, z) = \max_{0 \leq q_1 \leq (-z)} F_{t+1}(q_1 - (-y), (-z) - q_1)$. By antimultimodularity of F_{t+1} , $F_{t+1}(q_1 - (-y), (-z) - q_1)$ is supermodular in $(-y, q_1, -z)$ by (Hajek, 1985, Proposition 2.2). By preservation of the supermodularity by maximization, $H_{t+1}(y, z)$ is supermodular in $(-y, -z)$ and hence in (y, z) .

We are now ready to establish the sensitivity w.r.t. x_0 and a . Relation (12) can be rewritten as

$$\max_{a+z \geq 0, z \leq 0} V(z, x_1, x_0, a) \quad (13)$$

where $V(z, x_1, x_0, a) = h_t(x_0 + a + z) + H_{t+1}(x_1, z)$ and let z^* be the greatest maximizer. By concavity in z of $h_t(y)$ and $H_{t+1}(y, z)$, $V(z, x_1, x_0, a)$ is antimultimodular in (x_0, z) and in (a, z) . By (Li & Yu, 2014; Theorem 1), $-1 \leq \Delta_a z^* \leq 0$, $-1 \leq \Delta_{x_0} z^* \leq 0$. By $q_0 = (a + z^*) \wedge (s_{0t} - x_0)^+$, $0 \leq \Delta_a q_0 \leq 1$, $-1 \leq \Delta_{x_0} q_0 \leq 0$. From Theorem 4 and $q_1 + q_2 = -z^*$, $0 \leq \Delta_a q_i \leq 1$, $0 \leq \Delta_{x_0} q_i \leq 1$ for $i = 1$ and 2. (ii) is from $(\Delta_{x_0} z^* = 0 \rightarrow \Delta_{x_0}(x_0 + q_0) = 1, \Delta_{x_0}(q_1 + q_2) = 0)$ and $(\Delta_{x_0} z^* = -1 \rightarrow \Delta_{x_0}(x_0 + q_0) = 0, \Delta_{x_0}(q_1 + q_2) = 1)$. Similarly, $0 \leq \Delta_a(\sum_{i=0}^2 q_i) \leq 1$. The monotonicity in a of $\Delta_a(\sum_{i=0}^2 q_i)$ holds since $\Delta_a(\sum_{i=0}^2 q_i) = 0$ implies: (a) $x_0 + a + z^*(a) \geq s_{0t}$ due to $\Delta_a q_0 = 0$ and $\Delta_a z^* = 0$, (b) $z^*(a)$ maximizes $H_{t+1}(x_1, z)$ due to $\Delta_a q_0 = 0$ and the concavity in z of $H_{t+1}(x_1, z)$, and (c) $z^*(a') = z^*(a)$ for all $a' > a$.

We then establish the sensitivity w.r.t. x_1 . Assume first $0 \leq \Delta_{x_1} z^* \leq 1$. If $\Delta_{x_1} z^* = 0$, then $\Delta_{x_1} q_0 = 0$ and by Theorem 4, either $\Delta_{x_1} q_1 = \Delta_{x_1} q_2 = 0$ or $\Delta_{x_1} q_1 = -1, \Delta_{x_1} q_2 = 1$. If $\Delta_{x_1} z^* = 1$, then $\Delta_{x_1} q_0 \in \{0, 1\}$ and by Theorem 4, $\Delta_{x_1} q_1 = -1, \Delta_{x_1} q_2 = 0$ if $X_1(0, x_1 - z^*) \geq x_1 + 1$ and otherwise, no patient is assigned to q_1 and $\Delta_{x_1} q_1 = 0, \Delta_{x_1} q_2 = -1$. If the assumption $0 \leq \Delta_{x_1} z^* \leq 1$ is true, then so are all the sensitivity properties w.r.t. x_1 in (i) and (iv).

We now prove $0 \leq \Delta_{x_1} z^* \leq 1$. By properties of $H_{t+1}(y, z)$, $V(z, x_1, x_0, a)$ is pointwise concave and supermodular in (x_1, z) . The property $0 \leq \Delta_{x_1} z^*$ is from the concavity in z and

$$V(z^*, x_1 + 1, x_0, a) - V(z^* - 1, x_1 + 1, x_0, a) \geq V(z^*, x_1, x_0, a) - V(z^* - 1, x_1, x_0, a) \geq 0 \quad (14)$$

where the last inequalities of (14) is from the definition of z^* , the first inequality of (14) from the supermodularity in (x_1, z) . $\Delta_{x_1} z^* \leq 1$ follows from the concavity in z if

$$V(z^* + 2, x_1 + 1, x_0, a) - V(z^* + 1, x_1 + 1, x_0, a) < 0 \quad (15)$$

We now prove (15). From Theorem 4, $V(z, x_1, x_0, a) = h_t(x_0 + a + z) + H_{t+1}(x_1, z)$ and $H_{t+1}(x_1, z) = F_{t+1}(\bar{X}(x_1, -z))$. Let

$$A = \{(z^* + 2, x_1 + 1, x_0, a), (z^* + 1, x_1 + 1, x_0, a), (z^* + 1, x_1, x_0, a), (z^*, x_1, x_0, a)\}$$

Three cases are considered.

Case $X_1(0, x_1 - z^* - 1) \geq x_1 + 1$: $\bar{X}(x, -z) = \bar{X}(0, x - z)$ and $H_{t+1}(x, z) = H_{t+1}(0, z - x)$ for all four points in A and hence

$H_{t+1}(x_1 + 1, z^* + 2) - H_{t+1}(x_1 + 1, z^* + 1) = H_{t+1}(x_1, z^* + 1) - H_{t+1}(x_1, z^*)$ which together with the concavity of $h(x)$ and the definition of z^* implies (15).

Case $X_1(0, x_1 - z^*) \leq x_1 + 1$: $\bar{X}(x, -z) = (x, -z)$ and $H_{t+1}(x, z) = F_{t+1}(x, -z)$ for all four points in A . z^* is the maximal maximizer of $U(x, z) \triangleq h_t(x_0 + a + z) + F_{t+1}(x, -z)$ at $x = x_1$. $U(x, z)$ is antimultimodular in $(x, -z)$ and by (Li & Yu, 2014; Theorem 1), the maximal maximizer of $U(x, z)$ at $x = x_1 + 1$ is no more than $z^* + 1$. As a result,

$$V(z^* + 2, x_1 + 1, x_0, a) - V(z^* + 1, x_1 + 1, x_0, a) = U(x_1 + 1, z^* + 2) - U(x_1 + 1, z^* + 1) < 0$$

leading to (15).

Case $X_1(0, x_1 - z - 1) = x_1$ and $X_1(0, x_1 - z) = x_1 + 1$: $\bar{X}(x_1 + 1, -z^* - 2) = (x_1 + 1, -z^* - 2)$ and by definition, $V(z^* + 2, x_1 + 1, x_0, a) \leq W(x_1 + 1, z^* + 2)$ where $W(x, z) \triangleq h_t(x_0 + a + z) + H_{t+1}(0, z - x)$. For all other points in A , $H_{t+1}(x, z) = H_{t+1}(0, z - x)$ and $V(x, z) = W(x, z)$. As a result,

$$V(z^* + 2, x_1 + 1, x_0, a) - V(z^* + 1, x_1 + 1, x_0, a) \leq W(x_1 + 1, z^* + 2) - W(x_1 + 1, z^* + 1) \leq W(x_1, z^* + 1) - W(x_1, z^*) < 0$$

where the last two inequalities can be proved as for the first case.

Proof of Theorem 6. Assume by contradiction that there exists some $q_j > (s_{jt} - x_j)^+$ for some j . Consider the history-dependent policy π such that $q_{jt}^\pi = q_{jt} - 1, q_{it}^\pi = q_{it}, \forall i \neq j$ and $\bar{q}_t^\pi = \bar{q}_t, \forall t > t$. By construction,

$$f_t^\pi(\bar{x}) - f_t(\bar{x}) = E[-C_{t+j}(\Xi_{t+j} - 1) - r_j + C_{t+j}(\Xi_{t+j})] > 0$$

where Ξ_{t+j} is the total number of patients served in day $t+j$ if the optimal policy is applied and the inequality is from the convexity of $C_{t+j}(y)$ and $\Xi_{t+j} - 1 \geq x_j + q_j - 1 \geq s_{jt}$. $f_t^\pi(\bar{x}) - f_t(\bar{x}) > 0$ contradicts the assumption and concludes the proof.

Proof of Theorem 7. Assume by contradiction that $r_i - r_j > \max_{k>0} (r_{i-k} - r_{j-k})$ and $r_i - \Delta C_{t+i}(x_i + q_i) > r_j - \Delta C_{t+j}(x_j + q_j - 1)$ for some $i < j$ such that $q_j > 0$. Consider the history-dependent policy π such that moves a new patient from day $t+j$ to day $t+i$, i.e., $q_i^\pi = q_i + 1, q_j^\pi = q_j - 1$ and returns the next assignment from day $t+i$ to day $t+j$. As a result,

$$f_t^\pi(\bar{x}) - f_t(\bar{x}) = \sum_{k=1}^i P(W_k) (r_i + r_{j-k} - r_j - r_{i-k}) + P(U) E[r_i - \Delta C_{t+i}(x_i + q_i) - r_j + \Delta C_{t+j}(x_{t+j} - 1) | U]$$

where U is the event with no further assignment to day $t+i$, W_k is the event with at least one assignment to day $t+i$ on day $t+k$. By definition and the convexity of $C_{t+j}(y)$, $X_{t+j} \geq x_j + q_j$ and $\Delta C_{t+j}(X_{t+j} - 1) \geq \Delta C_{t+j}(x_j + q_j - 1)$. Since $\sum_{k=1}^i P(W_k) + P(U) = 1$, $f_t^\pi(\bar{x}) > f_t(\bar{x})$ which contradicts the optimality of $f_t(\bar{x})$ and concludes the proof.

Proof of Theorem 8. With the stationary reward, $s_{it} = s_{0,t+i}$. By Theorem 6, $q_i \leq (s_{0,t+i} - x_i)^+$ and $q_0 + \dots + q_L \leq a_t \wedge Z(\bar{x})$. Assume that the optimal control is such that $q_0 + \dots + q_L < a_t$ and $q_j < (s_{0,t+j} - x_j)^+$ for some j . Consider the history-dependent policy π such that adds a new patient to day $t+j$ and rejects the next patient assigned to day $t+j$ if any. As a result,

$$f_t^\pi(\bar{x}) - f_t(\bar{x}) = P(U) (r - \Delta C_{t+j}(x_j + q_j)) \geq 0$$

where U is the event with no further assignment to day $t+j$ and the inequality is from the convexity of $C_{t+j}(y)$ and $x_j + q_j < s_{jt}$. As a result, $f_t^\pi(\bar{x}) \geq f_t(\bar{x})$ and π is optimal. Repeating the above procedure leads to optimal control fulfilling the conditions of the theorem.

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